

RECONSTRUCTION-BASED ANOMALY DETECTION ON MVTEC AD: A COMPARISON OF AUTOENCODERS AND VARIATIONAL AUTOENCODERS

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Industrial visual inspection often requires detecting rare, unknown defects from mostly normal data. In this report, we study reconstruction-based anomaly detection using two unsupervised models: a convolutional Autoencoder (AE) and a Variational Autoencoder (VAE). Both models are trained only on normal (“good”) images and anomalies are detected via reconstruction errors and derived anomaly maps. We evaluate the methods on multiple classes of the MVTEC Anomaly Detection dataset using image-level and pixel-level metrics (AUROC, pixel AUPR) and complementary reconstruction quality indicators (PSNR, SSIM, LPIPS). Beyond quantitative results, we provide qualitative panels (original/reconstruction/GT mask/anomaly map/error map), loss curves, and score distribution plots to analyze failure modes, especially on texture categories where accurate reconstruction is challenging. Our experiments show that the plain AE is generally more reliable than the VAE in our setting, while the VAE tends to produce overly smooth reconstructions that can deteriorate localization on textures.

Index Terms—Anomaly Detection, Industrial Inspection, MVTEC AD, Autoencoder, Variational Autoencoder, Reconstruction Error, AUROC, AUPR, Localization.

I. INTRODUCTION

Industrial visual inspection systems are increasingly deployed to detect defective products in manufacturing pipelines. In many real settings, defective samples are rare, diverse, and sometimes unknown in advance. This makes supervised learning difficult because collecting and labeling representative anomalies is expensive and incomplete. As a result, unsupervised and one-class anomaly detection methods, trained only on normal data, have become a practical solution for defect detection.

A popular family of approaches is *reconstruction-based anomaly detection*. The core idea is simple: train a model to reconstruct normal images accurately. At test time, an image that contains an anomaly should be reconstructed poorly in the anomalous regions, yielding a high reconstruction error. This error can be aggregated into an *image-level anomaly score* (to classify a sample as normal or anomalous) and also projected spatially into a *pixel-level anomaly map* (to localize defects).

In this report, we compare two reconstruction models: a standard convolutional Autoencoder (AE) and a Variational Autoencoder (VAE). The VAE introduces a probabilistic latent space with a Kullback–Leibler (KL) regularization term that encourages smoothness and a structured representation. In principle, this regularization may improve generalization and robustness. However, it can also lead to overly smooth reconstructions that remove fine details, which is critical in industrial textures. Therefore, it is not obvious whether the VAE is beneficial for anomaly detection and localization compared to a plain AE.

We conduct experiments on several object and texture classes from the MVTEC Anomaly Detection (MVTEC AD) dataset. We evaluate both models using:

- **Image-level detection:** AUROC based on anomaly scores.

- **Pixel-level localization:** AUROC and AUPR computed from anomaly maps against ground-truth masks.
- **Reconstruction quality:** PSNR, SSIM, and LPIPS to characterize how reconstructions differ from original images.
- **Qualitative analysis:** visualization panels including original images, reconstructions, ground-truth masks, anomaly maps, and reconstruction error maps.

The report is organized as follows:

- Section II reviews anomaly detection approaches for industrial inspection, with a focus on reconstruction-based methods and common evaluation metrics.
- Section III describes our AE/VAE models, training protocol, anomaly scoring strategy, and evaluation pipeline.
- Section IV presents quantitative and qualitative results, including per-class comparisons and failure case analysis (notably for texture categories).
- Section V summarizes the main findings and outlines possible improvements.

II. RELATED WORK

This section summarizes the main paradigms used for industrial visual anomaly detection and details the evaluation protocol typically adopted on MVTEC AD. We intentionally keep the scope focused on families that directly help interpret our results with Autoencoders (AE) and Variational Autoencoders (VAE), while briefly positioning stronger modern alternatives.

A. Industrial visual anomaly detection and MVTEC AD

Industrial anomaly detection aims at identifying products or surfaces that deviate from normal manufacturing patterns, such as scratches, cracks, contaminations, missing parts, or structural deformations. In real factories, anomalous data are scarce, highly diverse, and often unknown in advance. For this

reason, many approaches adopt a *one-class* or *semi-supervised* setting: the model is trained mainly (or only) on *normal* samples and must detect deviations at test time.

A widely-used benchmark for this task is the **MVTec Anomaly Detection (MVTec AD)** dataset, composed of multiple categories split into *textures* (e.g., carpet) and *objects* (e.g., bottle, hazelnut, screw, grid). For each category, training data contain only normal images (`train/good`), while the test set (`test`) includes both normal and defective images. Importantly, defective test images are associated with **pixel-level ground-truth masks** (`ground_truth`), enabling both *detection* (image-level decision) and *localization* (pixel-level anomaly maps).

B. Reconstruction-based anomaly detection

A classical and intuitive strategy is to learn how *normality looks like* and flag deviations as anomalies. In **reconstruction-based** methods, a model is trained to reconstruct normal images. At test time, anomalies are expected to be poorly reconstructed, yielding large reconstruction errors that can be used as anomaly evidence.

a) **Autoencoders (AE)**.: An **Autoencoder** is a deterministic encoder–decoder network trained to minimize a reconstruction loss between the input image x and its reconstruction $\hat{x}$. Common losses include ℓ_1 (mean absolute error) or ℓ_2 (mean squared error). AEs are appealing as strong baselines: they are simple, fast, and often provide good reconstructions on normal samples. Their anomaly score is usually derived from a pixel-wise error map $e(x) = |x - \hat{x}|$ (or $(x - \hat{x})^2$), optionally smoothed and aggregated.

b) **Variational Autoencoders (VAE)**.: A **Variational Autoencoder** introduces a probabilistic latent space: the encoder predicts a distribution $q_\phi(z|x)$ and the decoder samples z to reconstruct $\hat{x}$. Training optimizes a sum of (i) a reconstruction term and (ii) a regularization term that enforces closeness to a prior $p(z)$ through the **Kullback–Leibler (KL) divergence**. In practice, VAEs can yield smoother latent representations and better generalization, but they may produce *blurrier reconstructions*, especially for high-frequency textures, which can affect localization maps.

c) **Common strengths and failure modes**.: Reconstruction-based approaches are conceptually straightforward and require only normal training data. However, two known limitations are central when interpreting results:

- **Over-reconstruction of anomalies**: if the model has enough capacity or if anomalies are visually close to normal patterns, it may reconstruct anomalies well, reducing reconstruction error and harming detection.
- **Texture reconstruction difficulty**: highly textured categories can be intrinsically hard to reconstruct precisely; reconstruction error may become high even for normal regions, producing noisy anomaly maps and false positives.

These limitations motivate the comparison of AE and VAE as representative reconstruction baselines, and justify why reconstruction quality metrics alone do not fully predict anomaly detection performance.

C. Feature-based and patch-based approaches (context)

To overcome reconstruction limitations, many state-of-the-art methods rely on features extracted from pretrained deep networks. The main idea is that normality can be modeled in a *feature space* rather than in pixel space:

- **Nearest-neighbor patch matching**: embeddings of normal patches are stored in a memory bank. At test time, each patch is scored by its distance to the nearest normal neighbor, producing accurate localization maps.
- **Distribution modeling in feature space**: normal embeddings are modeled with parametric or non-parametric distributions; anomalies are detected as low-likelihood or high-distance observations.

These approaches often outperform plain AE/VAE on MVTec AD, but they introduce additional design choices (feature extractor, layers, patch sizes, sampling, memory bank size, distances) and may be heavier computationally. In this report, we deliberately focus on AE and VAE to analyze reconstruction-based behavior and typical failure cases, using modern feature-based methods only as contextual reference.

D. Evaluation metrics on MVTec AD

MVTec AD evaluation typically distinguishes **image-level detection** from **pixel-level localization**. In our experiments, we report both.

1) Image-level metrics

a) **AUROC (Area Under the Receiver Operating Characteristic)**.: At image level, each test image is assigned a single anomaly score $s(x)$ (e.g., the average or maximum reconstruction error, or a robust aggregation). The **AUROC** measures how well the score separates normal vs defective images *across all thresholds*. An AUROC of 0.5 corresponds to random ranking, and 1.0 indicates perfect separation. A higher AUROC is better.

2) Pixel-level metrics

a) **Pixel-wise AUROC**.: For localization, the model outputs an anomaly map $A(x) \in \mathbb{R}^{H \times W}$ that is compared to the ground-truth defect mask $M(x) \in \{0, 1\}^{H \times W}$. Pixel AUROC evaluates ranking quality across all pixels. It is useful but may be overly optimistic in strongly imbalanced settings.

b) **Pixel-wise AUPR (Area Under the Precision–Recall Curve)**.: Because defect pixels are typically rare compared to background pixels, the dataset is highly imbalanced at pixel level. In such cases, the **AUPR** is often more informative than AUROC: it focuses on the precision/recall trade-off and is sensitive to false positives. A higher AUPR is better, but it can still be influenced by post-processing choices and thresholding.

3) Reconstruction quality metrics (interpretation tools)

In addition to detection/localization metrics, we report reconstruction fidelity metrics to better interpret failure modes:

a) **PSNR (Peak Signal-to-Noise Ratio)**.: PSNR is a logarithmic measure derived from the mean squared error (MSE) between x and $\hat{x}$. Higher PSNR indicates closer pixel-wise reconstruction. However, a high PSNR can be misleading for anomaly detection: the model might reconstruct anomalies too well and still obtain high PSNR while failing at localization.

b) *SSIM (Structural Similarity Index)*.: SSIM compares local luminance, contrast, and structure, aiming to better correlate with perceived image quality than pure MSE. Higher SSIM indicates more structurally faithful reconstructions. Like PSNR, SSIM does not guarantee good anomaly localization.

c) *LPIPS (Learned Perceptual Image Patch Similarity)*.: LPIPS measures perceptual similarity using deep feature embeddings. Lower LPIPS indicates reconstructions closer to the input in a perceptual sense. This metric is particularly relevant when reconstructions are blurry: an image can have moderate PSNR/SSIM but poor perceptual similarity, or vice versa.

d) *Key takeaway*.: Detection/localization performance (AUROC/AUPR) evaluates the anomaly scoring mechanism, while PSNR/SSIM/LPIPS mainly evaluate reconstruction faithfulness. In reconstruction-based anomaly detection, **better reconstructions are not always better for detection**: if anomalies are reconstructed too well, the anomaly map becomes weak. Therefore, reporting both families of metrics helps distinguish between (i) reconstruction limitations and (ii) anomaly scoring limitations.

III. METHODOLOGY

In this section, we describe our complete pipeline: data preparation, AE/VAE models, training protocol, anomaly scoring, and evaluation. The implementation follows a per-category setting on MVTec AD.

A. Problem setting

Let $\mathcal{D}_{train}$ be the set of normal (good) training images from one MVTec AD category. Our objective is to learn a reconstruction model using only $\mathcal{D}_{train}$ such that for a test image x we can produce:

- an **image-level anomaly score** $s(x)$ for detection (good vs defective),
- a **pixel-level anomaly map** $A(x)$ for localization.

B. Data preparation

All images are resized to 256×256 and processed as RGB inputs ($C = 3$). We split the normal training set into training/validation subsets to enable early stopping. Unless stated otherwise, preprocessing is identical for AE and VAE to ensure a fair comparison. We use a batch size $B = 16$ and train for up to 100 epochs.

C. Autoencoder (AE)

Our AE is a convolutional encoder–decoder network. Given an input x , the reconstruction is:

$$\hat{x} = D_{\phi}(E_{\theta}(x)).$$

The AE bottleneck is spatial, with 64 latent channels (AE_LATENT_CH= 64 in our configuration). The AE is trained by minimizing an ℓ_1 reconstruction loss:

$$\mathcal{L}_{AE}(x, \hat{x}) = \|x - \hat{x}\|_1.$$

We use ℓ_1 because it generally preserves edges better than ℓ_2 and often yields sharper reconstructions.

D. Variational Autoencoder (VAE)

The VAE encoder outputs $\mu(x)$ and $\log \sigma^2(x)$ that define an approximate posterior $q(z|x)$. We sample with the reparameterization trick:

$$z = \mu(x) + \sigma(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The decoder reconstructs $\hat{x} = D_{\phi}(z)$. We use a latent vector of dimension 256 (VAE_LATENT_DIM= 256). The training objective is:

$$\mathcal{L}_{VAE}(x, \hat{x}) = \mathcal{L}_{rec}(x, \hat{x}) + \beta \text{KL}(q(z|x) \parallel p(z)),$$

with $p(z) = \mathcal{N}(0, I)$. In our implementation, β follows a **warm-up schedule** from 0 to $\beta_{\max} = 0.01$ over the first 20 epochs (BETA_WARMUP_EPOCHS= 20), which prioritizes reconstruction early and progressively enforces latent regularization. To stabilize optimization, we apply gradient clipping with norm 5.0 (GRAD_CLIP= 5.0).

E. Training protocol

For each category, AE and VAE are trained only on normal images. We use Adam with learning rate 10^{-4} (for both models), weight decay 10^{-5} , and early stopping with patience $p = 10$ epochs. We keep the checkpoint with the best validation loss. Training curves (train/val loss vs epoch) are saved for each category. For the VAE, we also track the reconstruction loss and the KL term to analyze the effect of β warm-up.

F. Anomaly map and anomaly score

Given a test image x and reconstruction $\hat{x}$, we define the pixel-wise error:

$$E(x) = |x - \hat{x}|.$$

We reduce channels to obtain a 2D anomaly map:

$$A(x) = \text{mean}_c(E(x)).$$

For visualization, we optionally apply light spatial smoothing and normalization; metric computation always uses the same deterministic post-processing for comparability across models.

The image-level anomaly score aggregates $A(x)$. In our runs, we use the **99th percentile** (ANOM_SCORE = “p99”):

$$s(x) = \text{p99}_u(A_u(x)),$$

which increases sensitivity to small localized defects compared to the mean. (Alternative aggregations such as mean or max are possible, but are not used in our reported results.)

G. Evaluation

We evaluate on the test split:

- **Image AUROC** computed from $\{s(x_i)\}$ and image labels.
- **Pixel AUROC** and **Pixel AUPR** computed from anomaly maps and ground-truth masks.

Because defective pixels are rare, pixel AUPR is particularly informative. We also report reconstruction quality metrics (PSNR, SSIM, LPIPS) on test images to interpret reconstruction fidelity versus anomaly localization performance.

H. Pipeline pseudo-code

Algorithm 1: Reconstruction-based anomaly detection (AE/VAE)

Input: Normal train images $\mathcal{D}_{train}$, test images $\mathcal{D}_{test}$ with labels and masks

Output: Image scores $s(\cdot)$, anomaly maps $A(\cdot)$, evaluation metrics

- 1 Train AE/VAE on $\mathcal{D}_{train}$ minimizing reconstruction (and KL for VAE);
 - 2 **foreach** $x \in \mathcal{D}_{test}$ **do**
 - 3 Compute reconstruction $\hat{x}$;
 - 4 Compute pixel error $E(x) = |x - \hat{x}|$;
 - 5 Build anomaly map $A(x) = \text{mean}_c(E(x))$;
 - 6 Compute image score $s(x) = \text{p99}(A(x))$;
 - 7 Compute image AUROC from $\{s(x)\}$ and labels;
 - 8 Compute pixel AUROC/AUPR from $\{A(x)\}$ and masks;
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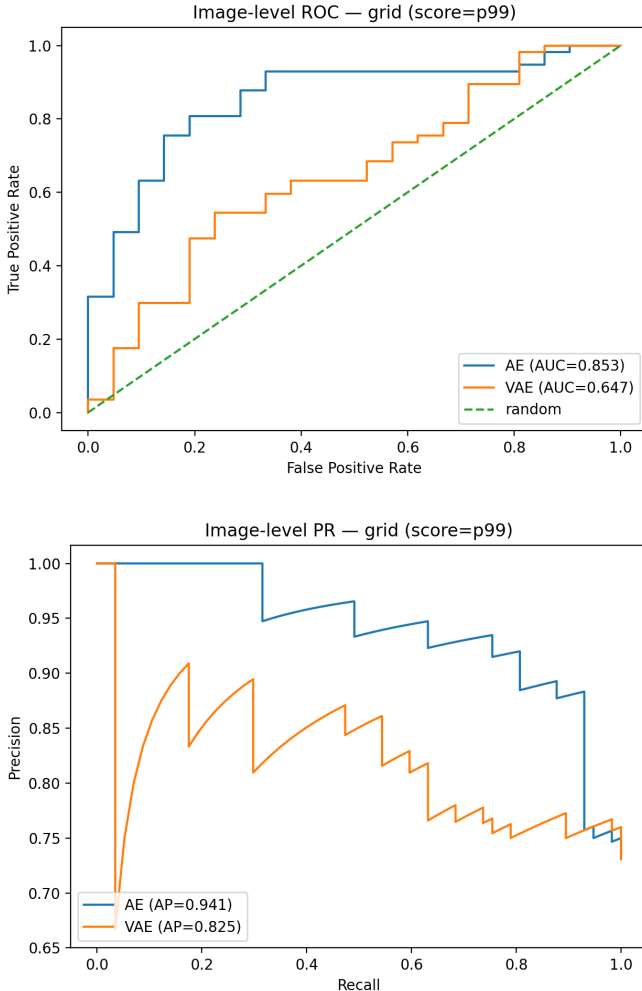


Fig. 1: Image-level ROC and Precision–Recall curves on *grid* for AE vs VAE (scores aggregated with p99).

IV. EXPERIMENTAL RESULTS

This section presents results on multiple MVTec AD categories, combining quantitative metrics (CSV summary) and qualitative analysis (loss curves, reconstruction panels, anomaly maps, and score distributions).

A. Experimental setup

For each class, AE and VAE are trained on the train/good split only. Evaluation is performed on the complete test split (good + anomalous) with pixel-level ground-truth masks when available. We report image-level AUROC, pixel-level AUROC and AUPR, as well as reconstruction metrics (PSNR, SSIM, LPIPS). Figures include:

- training/validation loss curves for AE and VAE,
- VAE loss decomposition (reconstruction, KL, and β warm-up effect),
- qualitative panels: original / reconstruction / GT mask / anomaly map / reconstruction error map,
- anomaly score distributions (good vs anomalous).

B. Quantitative results

Table I reports the main metrics for five categories. Overall, the AE baseline achieves higher image-level AUROC on **4 out of 5** classes (screw, carpet, hazelnut, grid). The only exception is *bottle*, where the VAE improves image AUROC (0.7198 vs 0.6492). At pixel level, AE and VAE can be close on object categories (e.g., *bottle* pixel AUROC ≈ 0.852 for both), while on textures the VAE can degrade severely (e.g., *grid* pixel AUPR drops to 0.0106).

Reconstruction metrics help interpret these behaviors: on texture classes, the VAE shows extremely low SSIM and very high LPIPS (e.g., *carpet* SSIM 0.0815, LPIPS 0.8258; *grid* SSIM 0.1672, LPIPS 0.8719), indicating reconstructions that do not preserve the normal texture structure.

C. Training dynamics

Across categories, AE and VAE losses decrease rapidly during early epochs and then plateau, with train/val curves remaining close, suggesting limited overfitting under our early stopping strategy.

For the VAE, the KL term is strongly influenced by the β warm-up: it typically increases early as β grows, then stabilizes once the latent distribution and reconstructions reach a balance. Even with a small $\beta_{\max} = 0.01$, the VAE can favor smoother reconstructions, which is detrimental when normality depends on high-frequency patterns.

D. Qualitative analysis: object vs texture categories

Qualitative panels provide crucial insight beyond global metrics:

- **Object classes (bottle, hazelnut, screw):** anomalies are often localized structural defects. A good reconstruction preserves the normal object while failing on defective regions, yielding concentrated anomaly maps and meaningful localization.

Class	Model	AUROC (img)	AUROC (px)	AUPR (px)	PSNR	SSIM	LPIPS
screw	AE	0.5839	0.9495	0.0420	30.6548	0.9062	0.2748
screw	VAE	0.4987	0.9423	0.0330	27.6101	0.8883	0.2935
carpet	AE	0.5915	0.8081	0.0469	21.0514	0.6049	0.2312
carpet	VAE	0.3419	0.4104	0.0145	19.0001	0.0815	0.8258
hazelnut	AE	0.8507	0.9606	0.4408	31.0281	0.8714	0.2724
hazelnut	VAE	0.7382	0.9710	0.4454	29.0404	0.8676	0.2891
grid	AE	0.8530	0.9566	0.4641	25.7691	0.8497	0.0587
grid	VAE	0.6475	0.6456	0.0106	17.1727	0.1672	0.8719
bottle	AE	0.6492	0.8522	0.1964	28.9569	0.8793	0.1241
bottle	VAE	0.7198	0.8520	0.1963	26.7183	0.8513	0.1384

TABLE I: Quantitative summary (image AUROC, pixel AUROC, pixel AUPR, and reconstruction quality metrics) for AE vs VAE on several MVTec AD classes.

- **Texture classes (carpet, grid):** normality is defined by repeated high-frequency patterns. If the reconstruction is blurred or distorted, the error spreads over normal regions, producing false positives and poor localization. This failure mode is especially apparent for the VAE in our results (Figures ?? and 6).

E. Score distributions and separability

To interpret image AUROC values, we visualize the distribution of anomaly scores for good vs anomalous images (computed as the 99th percentile of the anomaly map). When the distributions overlap heavily, image AUROC is necessarily low, even if some pixel-level localization remains possible. Figure 7 illustrates such a case for *grid* with the VAE, where separability is limited.

F. Discussion of failure cases

Several categories highlight typical limitations:

- **Carpet (VAE):** extremely low SSIM (0.0815) and very high LPIPS (0.8258) indicate reconstructions far from the original texture. This causes large reconstruction errors even on normal regions, degrading both detection and localization (image AUROC 0.3419).
- **Grid (VAE):** the VAE produces overly smooth or mismatched repetitive patterns, resulting in poor localization (pixel AUROC 0.6456, pixel AUPR 0.0106) and low SSIM/high LPIPS.
- **Bottle:** the VAE improves image-level separability (image AUROC 0.7198), while pixel-level AUROC remains similar to the AE. This suggests that, for some object categories, latent regularization can help the image score discriminate defects without necessarily improving localization.

G. Takeaways

Under our implementation and hyperparameters:

- A plain AE is a strong and reliable baseline for reconstruction-based anomaly detection.
- The VAE is sensitive to textures and may require careful tuning (capacity, β schedule, perceptual losses, or multi-scale designs) to avoid overly smooth reconstructions.

- Qualitative panels are essential to verify whether good metrics correspond to meaningful localization or to reconstruction artifacts.

V. CONCLUSION

We compared a convolutional Autoencoder (AE) and a Variational Autoencoder (VAE) for reconstruction-based anomaly detection on five MVTec AD categories (*screw*, *carpet*, *hazelnut*, *grid*, *bottle*). Both models were trained exclusively on normal images and evaluated with image-level AUROC, pixel-level AUROC/AUPR, and reconstruction-quality metrics (PSNR, SSIM, LPIPS). Beyond numerical scores, we analyzed training curves, VAE loss components (reconstruction vs KL under β warm-up), anomaly score distributions (using the p99 aggregation), and qualitative panels (reconstructions and anomaly maps).

Overall, the AE appears as the most reliable baseline in our setting: it achieves higher image-level AUROC on 4 out of 5 categories and yields more faithful reconstructions, especially on texture-like classes. In contrast, the VAE—despite its theoretically appealing probabilistic latent space—tends to produce smoother reconstructions, which can severely harm texture fidelity. This failure mode is clearly reflected by reconstruction metrics (very low SSIM and very high LPIPS on *carpet* and *grid*) and by the corresponding poor localization performance (notably the very low pixel AUPR on *grid*). However, on some object categories the VAE remains competitive and can even improve image-level separability (e.g., *bottle*), suggesting that latent regularization may help detection in specific cases.

Several extensions could be explored in future work. First, the VAE could benefit from additional tuning of the latent regularization (alternative β schedules, capacity changes, or different priors). Second, reconstruction objectives could be strengthened with perceptual and/or multi-scale losses to better preserve textures. Finally, hybrid approaches combining reconstruction errors with feature-based distances (e.g., pre-trained embeddings) would likely reduce the “reconstructing anomalies too well” limitation. Despite its simplicity, this study highlights a key practical takeaway: reconstruction-only baselines are strongly conditioned by the visual nature of the category, and texture classes require much more care than object classes to avoid widespread false positives and degraded localization.

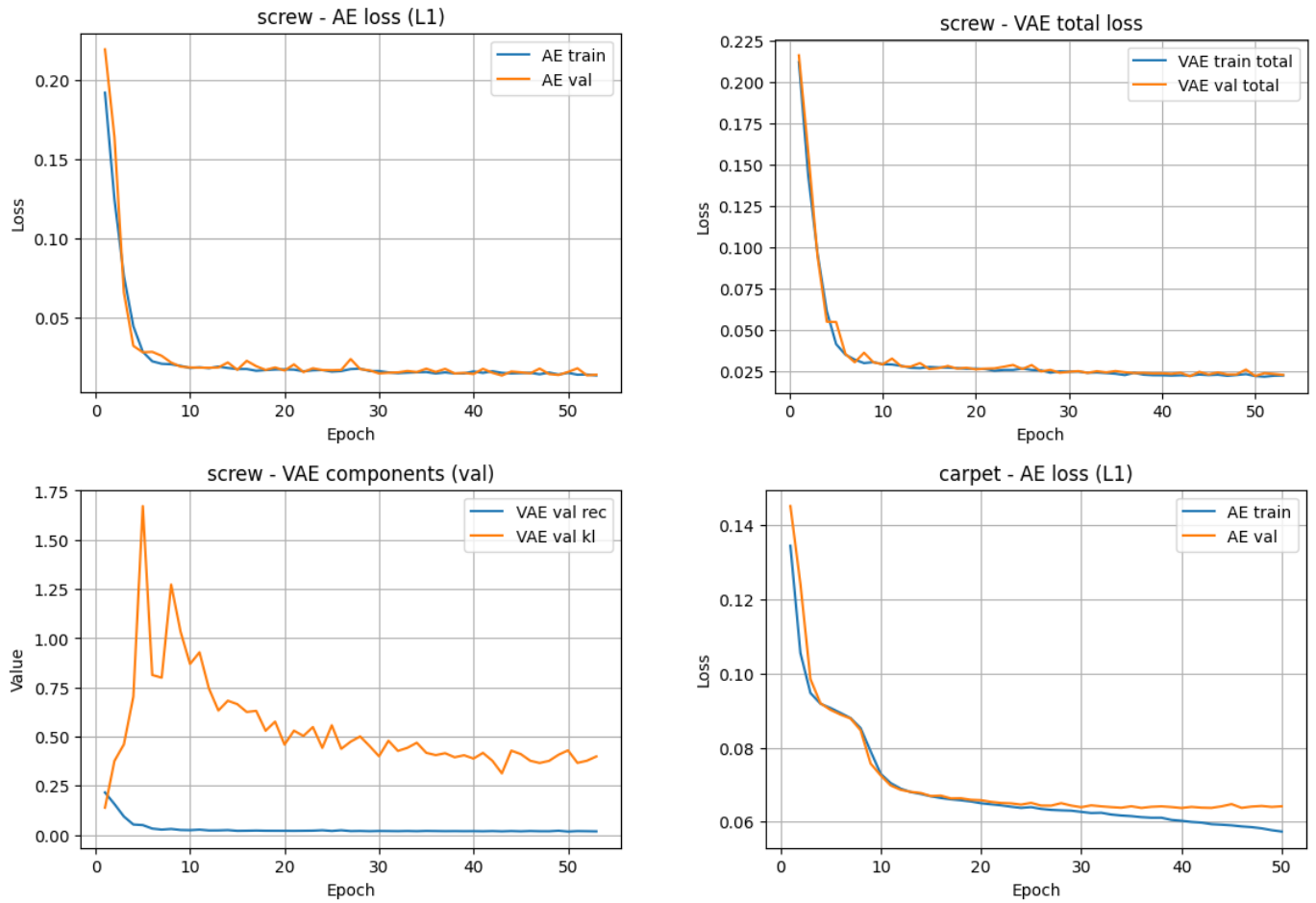


Fig. 2: Example training curves: AE loss (L1), VAE total loss, and VAE components (reconstruction vs KL). Similar plots are provided for each class.

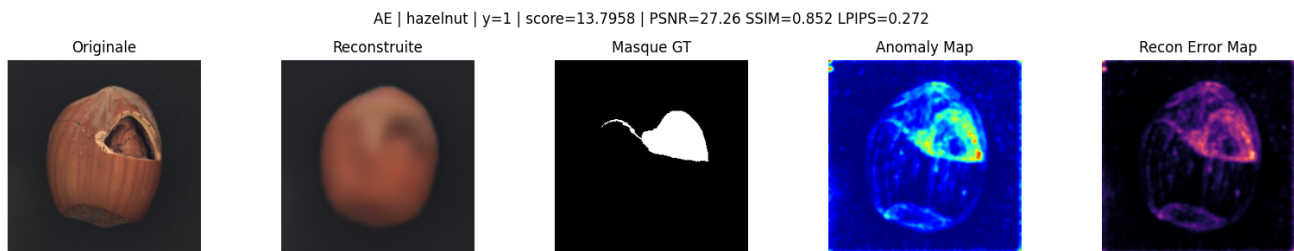


Fig. 3: Qualitative panel (AE) on *hazelnut*.

REFERENCES

APPENDIX A ADDITIONAL FIGURES

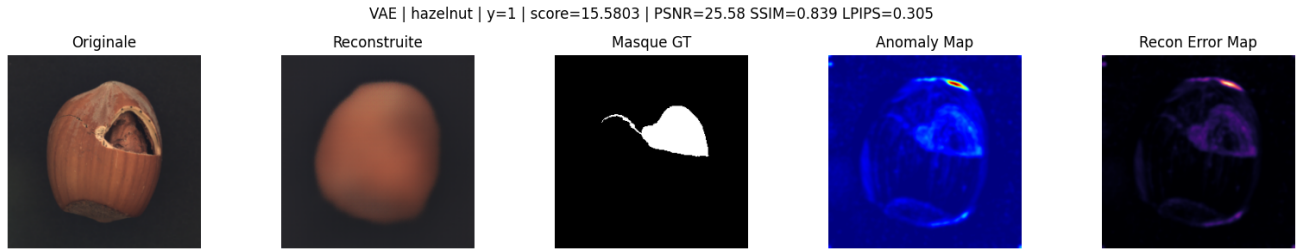


Fig. 4: Qualitative panel (VAE) on *hazelnut*.

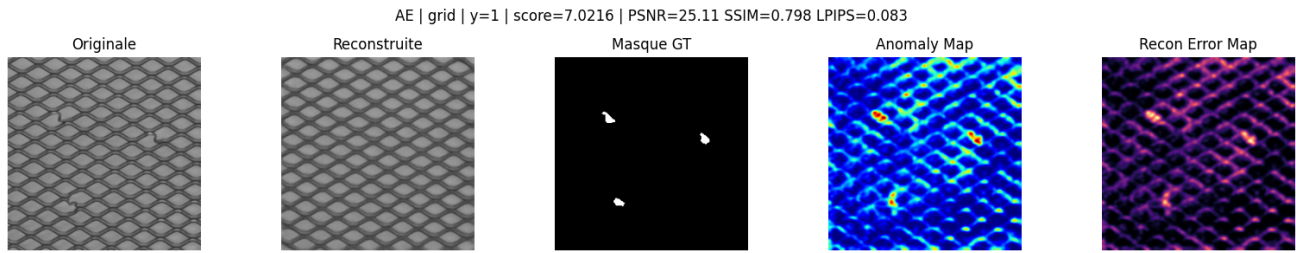


Fig. 5: Qualitative panel (AE) on *grid*.

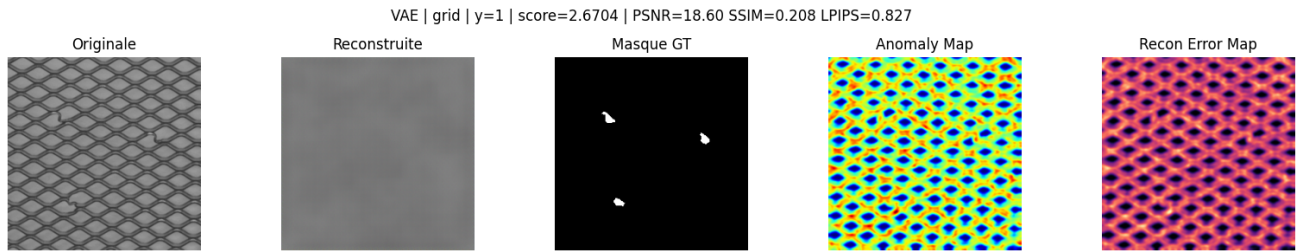


Fig. 6: Qualitative panel (VAE) on *grid*. Texture-like repetition often amplifies reconstruction artifacts, harming localization.



Fig. 7: Example score distribution (good vs anomalous) for *grid* with the VAE. Overlap indicates limited separability and correlates with lower image-level AUROC.

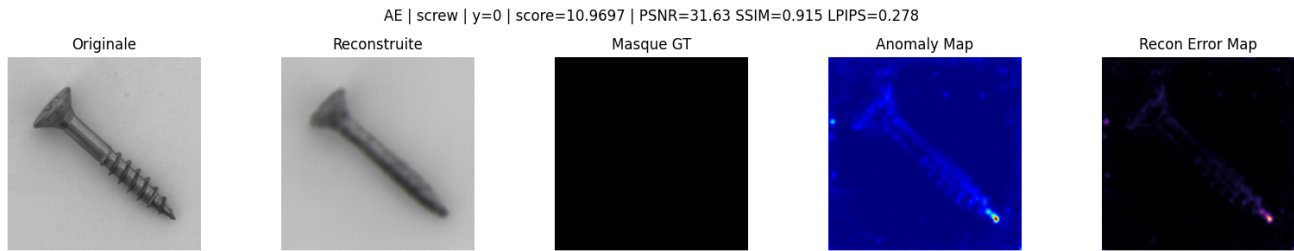


Fig. 8: Additional qualitative panel (AE) on *screw*: original, reconstruction, ground-truth mask, anomaly map, and reconstruction error map.

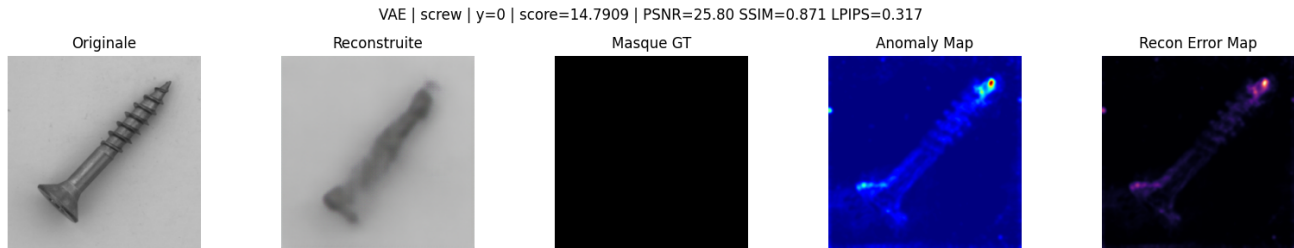


Fig. 9: Additional qualitative panel (VAE) on *screw*: original, reconstruction, ground-truth mask, anomaly map, and reconstruction error map.

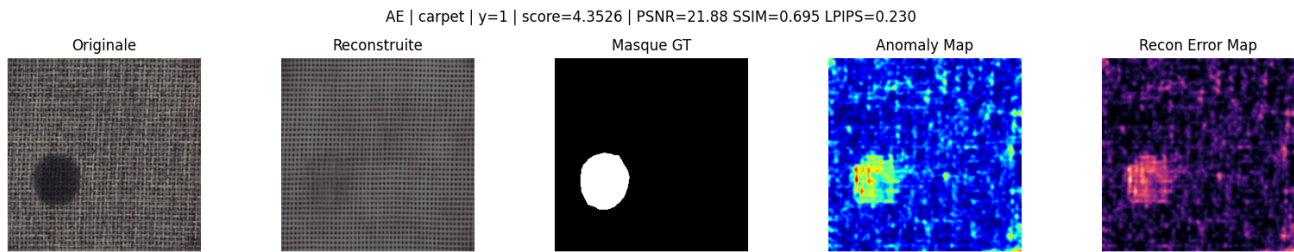


Fig. 10: Additional qualitative panel (AE) on *carpet*.

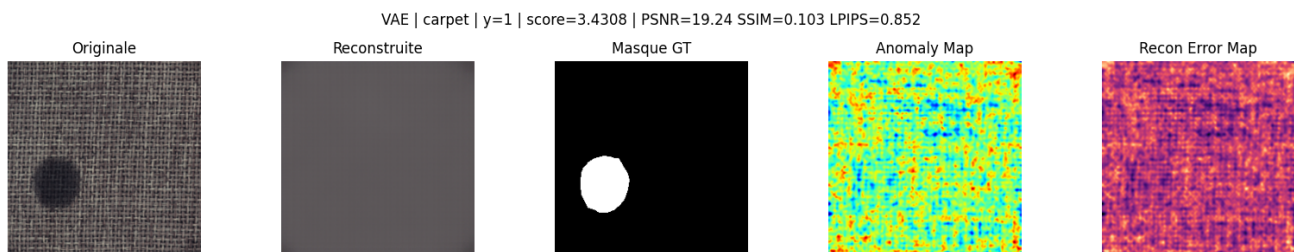


Fig. 11: Additional qualitative panel (VAE) on *carpet*. Texture reconstruction difficulty can lead to widespread errors and poor localization.

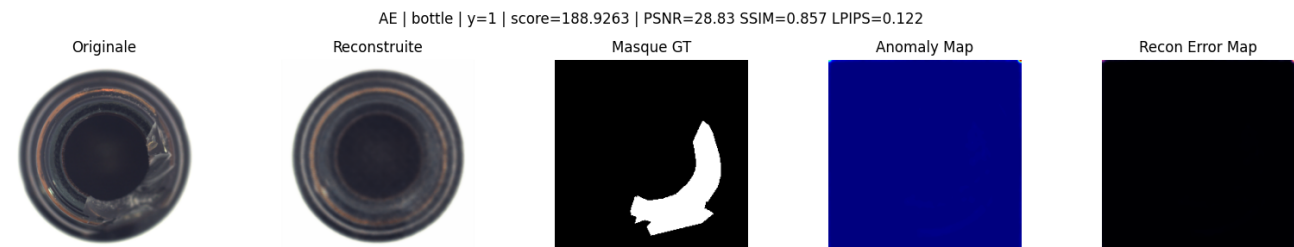


Fig. 12: Additional qualitative panel (AE) on *bottle*.

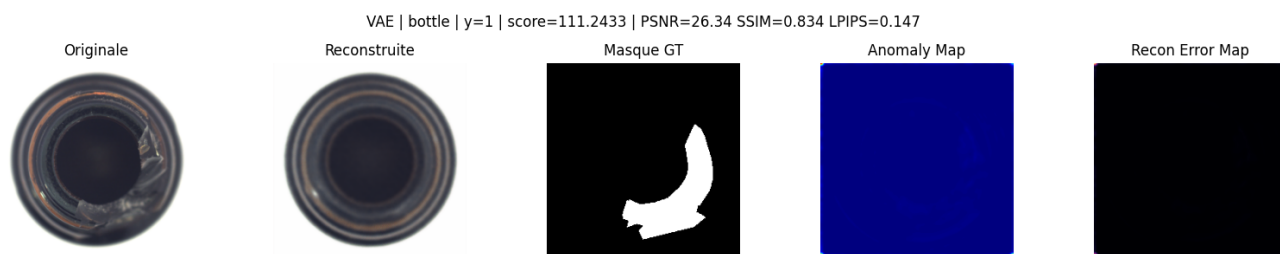


Fig. 13: Additional qualitative panel (VAE) on *bottle*.